CHAPTER 1 — COMPLEX NUMBERS

1. The Field of Complex Numbers

The section describes the system $\mathbb{C}$ of complex numbers and its operations including addition, multiplication, and division.

1.1. The Set $\mathbb{C}$ of Complex Numbers.

In order to find solutions of every quadratic equation, for example, $x^2 + 1 = 0$, the number system $\mathbb{R}$ of real numbers is not enough. We need to extend $\mathbb{R}$ to a new number system $\mathbb{C}$ by adding a new number i that is the square root of -1, that is, $i^2 = -1$. The set $\mathbb{C}$ is defined by

$$\mathbb{C} = \{z : z = x + iy \text{ with } x, y \in \mathbb{R}\}.$$

Each element $z \in \mathbb{C}$ is called a *complex number*. For $z = x + iy$, x is called the real part of z, denoted by $x = \operatorname{Re} z$ and y is called the imaginary part of z, denoted by $y = \operatorname{Im} z$.

The set $\mathbb{C}$ forms a complex plane. It has the following correspondence with the real plane $\mathbb{R}^2$,

$$\mathbb{C} \longrightarrow \mathbb{R}^2, \quad x + iy \longrightarrow (x, y).$$

A complex number z is viewed as a vector in the complex plane.

1.2. Addition and Multiplication.

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. The addition of complex numbers is defined by

Addition. $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$.

The addition can be illustrated by vector addition in the complex plane via parallelogram.

The multiplication of complex numbers is defined by

Multiplication. $z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_2 y_1 + x_1 y_2)$.

There are laws of operations:

$$z_1 + z_2 = z_2 + z_1, \quad z_1 z_2 = z_2 z_1 \quad \text{(commutative law)}$$

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), \quad (z_1 z_2) z_3 = z_1 (z_2 z_3) \quad \text{(associative law)}$$

$$z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3. \quad \text{(distributive law)}$$

We write $x + i0 = x$, and we have $z + 0 = z$, $0z = 0$ and $1z = z$.

1.3. Subtraction and Division.

Additive Inverse $-z$. The additive inverse of z is the complex number w so that $z + w = 0$. Obviously, $w = -z = -x + i(-y)$.

Subtraction.

$$z_1 - z_2 = z_1 + (-z_2) = (x_1 - x_2) + i(y_1 - y_2).$$

Multiplicative Inverse z^{-1}. The multiplicative inverse of z is the complex number w so that $zw = 1$.

Let $z = x + iy$ and $w = u + iv$. Then $zw = 1$ becomes

$$(xu - yv) + i(yu + xv) = 1 \quad \text{with solutions } u = \frac{x}{x^2 + y^2},\ v = \frac{-y}{x^2 + y^2},$$

when $x^2 + y^2 \neq 0$. Thus, when $z \neq 0$, we have

$$w = z^{-1} = \frac{x - iy}{x^2 + y^2} = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2} i. \tag{1.1}$$

Division.

$$\frac{z_1}{z_2} = z_1 z_2^{-1} = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} i, \quad \text{if } z_2 \neq 0.$$

Example.

$$\frac{1 + i}{2 - 3i} = (1 + i)(2 - 3i)^{-1} = (1 + i)\frac{2 + 3i}{13} = \frac{5 - i}{13} = \frac{5}{13} - \frac{1}{13} i.$$

By the definition of division, letting $z_1 = 1$ and $z_2 = z$ gives

$$\frac{1}{z} = z^{-1}, \quad z \neq 0.$$

We have

$$(z_1 z_2)^{-1} = z_1^{-1} z_2^{-1}, \quad z_1, z_2 \neq 0,$$

because

$$(z_1 z_2)(z_1^{-1} z_2^{-1}) = (z_1 z_1^{-1})(z_2 z_z^{-1}) = 1.$$

This identity can be written as

$$\frac{1}{z_1}\frac{1}{z_2} = \frac{1}{z_1 z_2}, \quad z_1, z_2 \neq 0.$$

Thus,

$$\frac{z_1}{z_2}\frac{z_3}{z_4} = \frac{z_1 z_3}{z_2 z_4}, \quad z_2, z_4 \neq 0,$$

because

$$\frac{z_1}{z_2}\frac{z_3}{z_4} = (z_1 z_2^{-1})(z_3 z_4^{-1}) = (z_1 z_3)(z_2^{-1} z_4^{-1}) = (z_1 z_3)(z_2 z_4)^{-1} = \frac{z_1 z_3}{z_2 z_4}.$$

There is also the identity

$$\frac{z_1}{z_2} = \frac{1}{z_2 z_1^{-1}},$$

because

$$\frac{z_1}{z_2} = z_1 z_2^{-1} = \frac{1}{(z_1 z_2^{-1})^{-1}} = \frac{1}{z_1^{-1} z_2}.$$

All the basic properties of fractions for real numbers hold for complex numbers.

Example.

$$\frac{1 + i}{2 - 3i} = \frac{(1 + i)(2 + 3i)}{(2 - 3i)(2 + 3i)} = \frac{5 - i}{13} = \frac{5}{13} - \frac{1}{13} i.$$

With the addition, multiplication and the additive and multiplicative inverses, the complex number system $\mathbb{C}$ is a field.

1.4. **Binomial Formula.**

$$(z_1 + z_2)^n = \sum_{k=0}^{n} \binom{n}{k} z_1^k z_2^{n-k}, \quad n = 1, 2, \ldots$$

2. Complex Conjugate and Modulus

Definition. Let z be a complex number. Then its complex *conjugate* $\bar{z}$ is defined by

$$\bar{z} = x - iy.$$

The *modulus* $|z|$ is defined by

$$|z| = (x^2 + y^2)^{\frac{1}{2}}.$$

Clearly, the modulus $|z|$ is just the norm, or length, of the vector z in the complex plane $\mathbb{C}$. We have

$$\operatorname{Re} z = x = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = y = \frac{z - \bar{z}}{2i}.$$

Then z is real if and only if $\bar{z} = z$, and z is imaginary if and only if $\bar{z} = -z$.

The following proposition can be shown easily.

Proposition 2.1. *Let $z, w \in \mathbb{C}$. Then*

(**a**) $\bar{\bar{z}} = z$;
(**b**) $z\bar{z} = |z|^2$;
(**c**) $\overline{z + w} = \bar{z} + \bar{w}$;
(**d**) $\overline{zw} = \bar{z}\bar{w}$;
(**e**) $|zw| = |z||w|$ *and* $|\bar{z}| = |z|$;
(**f**) $\max\{|Re\, z|, |Im\, z|\} \le |z| \le |Re\, z| + |Im\, z|$;
(**g**) $|z + w| \le |z| + |w|$. *(triangle inequality)*

The proof of the proposition is straightforward. The proof of $|zw| = |z||w|$:

Let $z = x + iy$ and $w = u + iv$. Then

$$|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = x^2u^2 + y^2v^2 + x^2v^2 + y^2u^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2|w|^2.$$

The proof of the triangle inequality:

$$\begin{aligned} |z + w|^2 &= (z + w)(\bar{z} + \bar{w}) = |z|^2 + 2\mathrm{Re}(z\bar{w}) + |w|^2 \le |z|^2 + 2|z\bar{w}| + |w|^2 \\ &= |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2. \end{aligned}$$

The multiplicative inverse z^{-1}, the complex conjugate $\bar{z}$ and the modulus $|z|$ have the relation,

$$z^{-1} = \frac{1}{z} = \frac{\bar{z}}{z\bar{z}} = \frac{\bar{z}}{|z|^2}. \tag{2.1}$$

This is the same formula in (1.1). Thus, $z^{-1} = \bar{z}$ if $|z| = 1$.

Example. (**1**) Prove the parallelogram law

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

We have

$$\begin{aligned}|z_1 + z_2|^2 + |z_1 - z_2|^2 &= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)\\ &= z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2\\ &= 2z_1\bar{z}_1 + 2z_2\bar{z}_2\\ &= 2(|z_1|^2 + |z_2|^2).\end{aligned}$$

(**2**) Let $c \in \mathbb{C}$. If $|z| = 1$ and $1 - \bar{c}z \neq 0$, show that

$$\frac{|z - c|}{|1 - \bar{c}z|} = 1.$$

Since $|z| = 1$, we have $\bar{z}z = 1$. Then

$$\frac{|z - c|}{|1 - \bar{c}z|} = \frac{|z - c|}{|\bar{z}z - \bar{c}z|} = \frac{|z - c|}{|\bar{z} - \bar{c}||z|} = \frac{|z - c|}{|z - c|} = 1.$$

(**3**) If $|z| = 3$, show that

$$\left|\frac{1}{z^2 - (1 + i)z + i}\right| < \frac{1}{4}.$$

We have $z^2 - (1 + i)z + i = (z - 1)(z - i)$ and $|z - 1| \geq |z| - 1 = 2$, $|z - i| \geq 2$. Then

$$\left|\frac{1}{z^2 - (1 + i)z + i}\right| = \frac{1}{|z^2 - (1 + i)z + i|} = \frac{1}{|z - 1||z - i|} < \frac{1}{4},$$

because $|z - 1| = 2$ and $|z - i| = 2$ not simultaneously.

3. Polar Form, Argument and Exponential Form

3.1. Polar Form and Argument.

Let $z = x + iy$ and let r, θ be the polar coordinates of (x, y), that is, $x = r\cos\theta$, $y = r\sin\theta$. Then the complex number z can be written as the *polar form*,

$$z = r(\cos\theta + i\sin\theta). \tag{3.1}$$

The radial coordinate r is equal to the modulus z, $r = |z|$. For $z \neq 0$, the angle coordinate θ has infinitely many values which are 2π periodic. Each value of θ is called an *argument* of z. The set of all arguments is denoted by $\arg z$. The *principal value* of $\arg z$, denoted by $\operatorname{Arg} z$, is the value of θ satisfying $-\pi < \theta \leq \pi$. Thus,

$$\arg z = \{\operatorname{Arg} z + 2k\pi : k = 0, \pm 1, \pm 2, \ldots\}, \quad z \neq 0. \tag{3.2}$$

By

$$\bar{z} = r(\cos\theta - i\sin\theta) = r(\cos(-\theta) + i\sin(-\theta)),$$

we have

$$\arg \bar{z} = -\arg z. \tag{3.3}$$

Example.

$$\operatorname{Arg} i = \frac{\pi}{2}, \quad \operatorname{Arg}(1-i) = -\frac{\pi}{4}, \quad \operatorname{Arg}(-1-i) = -\frac{3\pi}{4},$$

and

$$\arg i = \Big\{\frac{\pi}{2} + 2k\pi : k = 0, \pm 1, \pm 2, \dots\Big\}, \quad \arg(-1-i) = \Big\{-\frac{3\pi}{4} + 2k\pi : k = 0, \pm 1, \pm 2, \dots\Big\}.$$

Even though $\operatorname{Arg}(-1-i) \neq \frac{5\pi}{4}$, one can write

$$\arg(-1-i) = \Big\{\frac{5\pi}{4} + 2k\pi : k = 0, \pm 1, \pm 2, \dots\Big\}.$$

3.2. **Exponential Form.**

The notation $e^{i\theta}$ is defined by

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

which is called Euler's formula and is consistent with the exponential function to be defined later.

The *exponential form* of z is

$$z = re^{i\theta}, \quad r = |z|, \ \theta = \arg z. \tag{3.4}$$

Example.

$$-1-i = \sqrt{2}e^{i(-\frac{3\pi}{4})} = \sqrt{2}e^{i(-\frac{3\pi}{4})+2k\pi}, \quad k = 0, \pm 1, \pm 2, \dots$$

$$e^{i\pi} = -1, \quad e^{i\pi/2} = i, \quad e^{-i\pi/2} = -i, \quad e^{i\pi/3} = \frac{1+\sqrt{3}i}{2}, \quad e^{i\pi/4} = \frac{1+i}{\sqrt{2}}.$$

By using trigonometric formulas, it is easy to verify the following exponential law:

$$e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}. \tag{3.5}$$

$$\begin{aligned}(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2) &= \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\\ &= \cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2).\end{aligned}$$

Let $z_1 = r_1e^{i\theta_1}$ and $z_2 = r_2e^{i\theta_2}$. By the exponential law (3.5), it follows that

$$z_1z_2 = r_1r_2e^{i(\theta_1+\theta_2)}, \quad \frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1-\theta_2)}, \quad z_1^n = r_1^ne^{in\theta_1}, \quad n = 0, \pm 1, \pm 2, \dots$$

These formulas imply that

$$\arg(z_1z_2) = \arg z_1 + \arg z_2, \quad \arg\Big(\frac{z_1}{z_2}\Big) = \arg z_1 - \arg z_2. \tag{3.6}$$

and

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta, \quad n = 0, \pm 1, \pm 2, \dots$$

Example.

(**1**) Let $z_1 = -1$ and $z_2 = i$. Find $\operatorname{Arg}(z_1 z_2)$ and $\operatorname{Arg}\Big(\dfrac{z_1}{z_2}\Big)$.

$$\operatorname{Arg}(z_1 z_2) = \operatorname{Arg}(-i) = -\frac{\pi}{2}; \quad \operatorname{Arg}\Big(\frac{z_1}{z_2}\Big) = \operatorname{Arg}\Big(\frac{-1}{i}\Big) = \operatorname{Arg} i = \frac{\pi}{2}.$$

(**2**) Let $z = \dfrac{-2}{1+\sqrt{3}i}$. Find $\operatorname{Arg} z$.

Since $\operatorname{Arg}(-2) = \pi$ and $\operatorname{Arg}(1+\sqrt{3}i) = \dfrac{\pi}{3}$, by (3.6) one value of $\arg z$ is $\pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$ which is between $-\pi$ and π. Thus, $\operatorname{Arg} z = \dfrac{2\pi}{3}$.

Also, $z = -2 \cdot \dfrac{1-\sqrt{3}i}{4} = -\dfrac{1}{2} + \dfrac{\sqrt{3}}{2} i$, Thus, $\operatorname{Arg} z = \dfrac{2\pi}{3}$.

(**3**) Let $z = \dfrac{-2}{1-\sqrt{3}i}$. Find $\operatorname{Arg} z$.

Since $\operatorname{Arg}(-2) = \pi$ and $\operatorname{Arg}(1-\sqrt{3}i) = -\dfrac{\pi}{3}$, by (3.6) one value of $\arg z$ is $\pi + \dfrac{\pi}{3}$. Thus,

$$\operatorname{Arg} z = \pi + \frac{\pi}{3} - 2\pi = -\frac{2\pi}{3}.$$

Also, $z = -2 \cdot \dfrac{1+\sqrt{3}i}{4} = -\dfrac{1}{2} - \dfrac{\sqrt{3}}{2} i$, Thus, $\operatorname{Arg} z = -\dfrac{2\pi}{3}$.

4. Roots of Complex Numbers

Given a complex number $z_0 = r_0 e^{i\theta_0}$, its nth roots are solutions z to the equation $z^n = z_0$.

Note that two complex numbers $z_1 = r_1 e^{i\theta_1}$ and $z_1 = r_1 e^{i\theta_1}$ are equal, $z_1 = z_2$, if and only if they have the same modulus and the same arguments, that is,

$$r_1 = r_2 \quad \text{and} \quad \theta_1 = \theta_2 + 2k\pi, \quad k = 0, \pm 1, \pm 2, \ldots$$

Therefore, the equation $z^n = z_0$, that is, $r^n e^{in\theta} = r_0 e^{i\theta_0}$, implies that

$$r^n = r_0, \quad \text{and} \quad n\theta = \theta_0 + 2k\pi, \quad k = 0, \pm 1, \pm 2, \ldots$$

It follows that

$$z = r_0^{\frac{1}{n}} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)}, \quad k = 0, \pm 1, \pm 2, \ldots \tag{4.1}$$

This is a periodical sequence. The points lie on the circle of radius $r_0^{\frac{1}{n}}$ and centered at the origin. Two consecutive points have a radian distance (angle) $\dfrac{2\pi}{n}$.

Therefore, the distinct roots of z_0 are

$$c_k = r_0^{\frac{1}{n}} e^{i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)}, \quad k = 0, 1, 2, \ldots, n-1.$$

Consequently, the n distinct roots of $z_0 = 1$ are

$$\omega_n^k, \quad k = 0, 1, 2, \ldots, n-1,$$

where $\omega_n = e^{i\frac{2\pi}{n}}$. We have

$$c_k = c_0 \omega_n^k, \quad k = 0, 1, 2, \ldots, n-1.$$

The root $c_0 = r_0^{\frac{1}{n}} e^{i\frac{\theta_0}{n}}$, $-\pi < \theta_0 \le \pi$, is called the *principal root* of z_0.

The n roots of unit, $1, \omega_n, \omega_n^2, \ldots, \omega_n^{n-1}$ are equally spaced points on the unit circle, which are vertices of the regular n-polygon inscribed in the unit circle with 1 as a vertex.

Example.

(**1**) Find the cube roots $(-8i)^{\frac{1}{3}}$.

$$-8i = 8e^{i(-\frac{\pi}{2}+2k\pi)}; \quad c_k = 2e^{i(-\frac{\pi}{6}+\frac{2k\pi}{3})}, \quad k = 0, 1, 2.$$

$$\begin{aligned} c_0 &= 2e^{i(-\frac{\pi}{6})} = 2(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}) = \sqrt{3} - i, \\ c_1 &= 2e^{i\frac{\pi}{2}} = 2i, \\ c_2 &= 2e^{i\frac{7\pi}{6}} = 2(-\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}) = -\sqrt{3} - i. \end{aligned}$$

(**2**) Find the square roots $(\sqrt{3}+i)^{\frac{1}{2}}$.

$$\sqrt{3} + i = 2e^{i(\frac{\pi}{6}+2k\pi)}; \quad c_k = \sqrt{2}e^{i(\frac{\pi}{12}+k\pi)}, \ k = 0, 1;$$

$$c_0 = \sqrt{2}\Big(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\Big) = \frac{\sqrt{2}}{2}\Big(\sqrt{2+\sqrt{3}} + i\sqrt{2-\sqrt{3}}\Big) = -c_1,$$

which follows from

$$\begin{aligned} \cos\frac{\pi}{12} &= \Big(\frac{1}{2} + \frac{1}{2}\cos\frac{\pi}{6}\Big)^{\frac{1}{2}} = \Big(\frac{1}{2} + \frac{\sqrt{3}}{4}\Big)^{\frac{1}{2}}, \\ \sin\frac{\pi}{12} &= \Big(\frac{1}{2} - \frac{1}{2}\cos\frac{\pi}{6}\Big)^{\frac{1}{2}} = \Big(\frac{1}{2} - \frac{\sqrt{3}}{4}\Big)^{\frac{1}{2}}. \end{aligned}$$

5. Sequences of Complex Numbers

Sequence. A sequence of complex numbers is $z_1, z_2, \ldots, z_n, \ldots$, denoted by $\{z_n\}$.

5.1. **Convergence of a Sequence.**

A sequence of complex numbers $\{z_n\}$ is said to converge to the complex number z if for any $\varepsilon > 0$ there exits an integer N so that

$$|z_n - z| < \varepsilon \quad \text{whenever} \quad n > N.$$

We write

$$\lim_{n\to\infty} z_n = z \quad \text{or} \quad z_n \to z \text{ as } n \to \infty.$$

(Squeezing Principle) It is easily seen that z_n converges to z if and only if there exists a sequence of positive numbers ε_n so that

$$|z_n - z| \le \varepsilon_n \to 0.$$

Example. (**1**) $\lim_{n\to\infty} z^n = 0$ if $|z| < 1$.

$$|z^n - 0| \le \varepsilon_n \to 0, \quad \text{where } \varepsilon_n = |z|^n.$$

(**2**) $\lim_{n\to\infty} \frac{1}{z^n} = 0$ if $|z| > 1$.

Let $w = \frac{1}{z}$. Then $|w| < 1$ and $w^n \to 0$.

(**3**) $\lim_{n\to\infty} \frac{z^n}{3^n+1} = 0$ if $|z| < 3$.

$$\left|\frac{z^n}{3^n+1} - 0\right| \le \frac{|z|^n}{3^n+1} \le \frac{|z|^n}{3^n} \le \varepsilon_n \to 0, \quad \text{where } \varepsilon_n = \left(\frac{|z|}{3}\right)^n.$$

Proposition 5.1. *Let $z_n = x_n + iy_n$ and $z = x + iy$. Then the sequence of complex numbers $\{z_n\}$ converges to the complex number z if and only if $x_n \to x$ and $y_n \to y$.*

The proposition follows from applications of the following inequalities (see Proposition 2.1):

$$|x_n - x| \le |z_n - z|$$
$$|y_n - y| \le |z_n - z|$$
$$|z_n - z| \le |x_n - x| + |y_n - y|.$$

Obviously, $|z_n - z| \to 0$ implies $|x_n - x| \to 0$ and $|y_n - y| \to 0$ and vice versa.

Example. (**1**) Let $z_n = \frac{n}{n+1} + i\sin\left(\frac{\pi}{2} + \frac{1}{n}\right)$. Then $z_n \to 1 + i$ because $\frac{n}{n+1} \to 1$ and $\sin\left(\frac{\pi}{2} + \frac{1}{n}\right) \to 1$.

(**2**) If $z_n \to z$, then $|z_n| \to |z|$, but $\mathrm{Arg}(z_n)$ may not be convergent. Here is an example: Let

$$z_n = \frac{1-n}{1+n} + i\frac{(-1)^n}{n}. \quad \text{Then} \quad z_n \to -1. \quad \text{But} \quad \mathrm{Arg}(z_{2k}) \to \pi, \quad \mathrm{Arg}(z_{2k+1}) \to -\pi.$$

Proposition 5.2. *If $z_n \to z$ and $w_n \to w$, then $z_n + w_n \to z + w$ and $z_n w_n \to zw$.*

This proposition can be shown by using Proposition 5.1.

Bounded sequence. A sequence $\{z_n\}$ of complex numbers is said to be *bounded* if $|z_n| \le M$ for some positive constant M.

Subsequence. For any integer sequence $0 < n_1 < n_2 < \ldots < n_k < \ldots$, $\{z_{n_k}\}$ is called a *subsequence* of sequence $\{z_n\}$.

Bolzano-Weierstrass Theorem. Any bounded sequence of complex numbers has a convergent subsequence.

Proof. Let $\{z_n\}$ be a bounded sequence. Let $z_n = x_n + iy_n$. Since $\max\{|x_n|, |y_n|\} \le |z_n|$, both $\{x_n\}$ and $\{y_n\}$ are bounded. By the Bolzano-Weierstrass theorem for sequences of real numbers, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$. The subsequence $\{y_{n_k}\}$ is bounded and thus has a convergent subsequence $\{y_{n_{k_l}}\}$. Therefore, by Proposition 5.1, $\{z_{n_{k_l}}\}$ is a convergent subsequence. $\square$

Example. Show that the sequence

$$z_n = \frac{6z^n + 4}{1 + |z|^n}, \quad n = 1, 2, \dots, \quad z \in \mathbb{C},$$

has a convergent subsequence. Give a value of z so that the sequence has 100 convergent subsequences.

The sequence z_n is bounded. By the triangle inequality,

$$|z_n| \le \frac{6|z|^n}{1 + |z_n|^n} + \frac{4}{1 + |z|^n} < 10.$$

The Bolzano-Weierstrass theorem shows that the sequence has a convergent subsequence.

Let $z = e^{i\frac{\pi}{200}}$. Then $|z| = 1$ and

$$z_n = 3e^{i\frac{n\pi}{200}} + 2,$$

which is periodic and has at least 100 convergent subsequences.

5.2. **Cauchy Sequence.**

A sequence of complex numbers $\{z_n\}$ is a Cauchy sequence if for any $\varepsilon > 0$ there exists integer $N > 0$ so that

$$|z_n - z_m| < \varepsilon \quad \text{whenever } n, m \ge N.$$

Let $z_n = x_n + iy_n$. By the inequalities,

$$\max\{|x_n - x_m|, |y_n - y_m|\} \le |z_n - z_m| \le |x_n - x_m| + |y_n - y_m|,$$

it follows immediately that $\{z_n\}$ is a Cauchy sequence if and only if $\{x_n\}$ and $\{y_n\}$ are Cauchy sequences.

By the Cauchy criterion for sequences of real numbers and Proposition 5.1, we have the following Cauchy criterion foe sequences of complex numbers.

Cauchy Criterion. A sequence of complex numbers converges if and only if it is a Cauchy sequence.

6. Series of Complex Numbers

Series. A series of complex numbers is $z_1 + z_2 + \cdots + z_n + \cdots$, denoted by $\sum_{n=1}^{\infty} z_n$.

6.1. **Convergence of a Series.**

A series $\sum_{n=1}^{\infty} z_n$ is said to be convergent if its partial sum sequence $\{s_n\}$ is convergent, where $s_n = \sum_{k=1}^{n} z_k$. The limit of s_n is called the sum of the series.

Example. Consider the geometric series

$$\sum_{n=1}^{\infty} z^{n-1}, \quad |z| < 1.$$

The partial sum sequence

$$s_n = 1 + z + z^2 + \cdots + z^{n-1} = \frac{1 - z^n}{1 - z} \to \frac{1}{1 - z}, \quad |z| < 1.$$

Thus,

$$\sum_{n=1}^{\infty} z^{n-1} = \frac{1}{1 - z}, \quad |z| < 1.$$

Cauchy Criterion. A series $\sum_{n=1}^{\infty} z_n$ is convergent if and only if for any $\varepsilon > 0$ there exists an integer N so that

$$\Big| \sum_{k=n}^{m} z_k \Big| < \varepsilon \quad \text{whenever } m \geq n \geq N.$$

Proposition 6.1. *Let $z_n = x_n + iy_n$. Then $\sum_{n=1}^{\infty} z_n$ is convergent if and only if $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are convergent.*

The proof follows from Proposition 5.1 and $S_n = X_n + iY_n$, where

$$S_n = \sum_{k=1}^{n} z_k, \quad X_n = \sum_{k=1}^{n} x_k, \quad Y_n = \sum_{k=1}^{n} y_k.$$

Corollary 6.2. *(Necessary condition) If $\sum_{n=1}^{\infty} z_n$ is convergent, then it is necessary that $z_n \to 0$.*

It follows from the necessary conditions, $x_n \to 0$ and $y_n \to 0$, for the convergence of $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$. It also follows from the Cauchy criterion by letting $n = m$.

Corollary 6.3. *(Absolute convergence) If $\sum_{n=1}^{\infty} |z_n|$ is convergent, then $\sum_{n=1}^{\infty} z_n$ is convergent.*

The convergence of $\sum_{n=1}^{\infty} |z_n|$ and the inequalities $|x_n| \leq |z_n|$ and $|y_n| \leq |z_n|$ imply that both $\sum_{n=1}^{\infty} x_n$ and $\sum_{n=1}^{\infty} y_n$ are absolutely convergent.

Corollary 6.4. *(Comparison test) If $|z_n| \leq a_n$ and $\sum_{n=1}^{\infty} a_n$ is convergent, then $\sum_{n=1}^{\infty} z_n$ is convergent.*

The conditions imply that $\sum_{n=1}^{\infty} |z_n|$ is convergent.

Example. (**1**) Suppose $\sum_{n=1}^{\infty} |a_n|$ is convergent and $|z| \leq 1$. Then $\sum_{n=1}^{\infty} a_n z^n$ is convergent.

This is because $|a_n z^n| \leq |a_n|$ and the convergence of $\sum_{n=1}^{\infty} |a_n|$.

(**2**) Suppose a_n is bounded and $|z| < 1$. Then $\sum_{n=1}^{\infty} a_n z^n$ is convergent.

This is because $|a_n z^n| = |a_n||z^n| \leq M|z|^n$ and the convergence of $\sum_{n=1}^{\infty} |z|^n$.

(**3**) The series $\sum_{n=1}^{\infty} \frac{1+i\sqrt{n}}{n+in^2}$ converges absolutely.

$$\left|\frac{1+i\sqrt{n}}{n+in^2}\right| = \frac{|1+i\sqrt{n}|}{|n+in^2|} = \frac{(1+n)^{\frac{1}{2}}}{(n^2+n^4)^{\frac{1}{2}}} \leq \frac{2n^{\frac{1}{2}}}{n^2} = \frac{2}{n^{\frac{3}{2}}}, \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}} < \infty.$$

(**4**) The series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges absolutely when $|z| < 1$ and diverges when $|z| > 1$.

When $|z| < 1$, the series $\sum_{n=1}^{\infty} \frac{|z|^n}{n}$ of positive numbers is convergent, which can be shown by the ratio test. By the comparison test, the series $\sum_{n=1}^{\infty} \frac{z^n}{n}$ converges absolutely.

When $|z| > 1$, $\frac{z^n}{n}$ does not converge to 0 because $\frac{|z|^n}{n} \to \infty$.

(**5**) The series $\sum_{n=1}^{\infty} \frac{z^n}{n!}$ converges for any $z \in \mathbb{C}$.

It is because $\sum_{n=1}^{\infty} \frac{|z|^n}{n!}$ converges for any $z \in \mathbb{C}$, which can be shown by the ratio test.

References

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